Law Office Management System

System Title

System Title: Law Office Management System

Description: Clear, specific, and directly descriptive of the administrative processes and users involved in managing a law firm

Part B — System Requirements

1. System Purpose: To streamline law firm operations by managing client intake, tracking active legal cases, scheduling appointments and court hearings, organizing case documents, and managing billable hours.

2. Target Users: The system serves four primary user groups with distinct access levels and responsibilities:

• **Administrator / Managing Partner:** Manages system settings, user roles, firm-wide reporting.

• **Lawyer / Attorney:** Manages assigned cases, uploads legal documents, logs billable hours, and tracks court schedules.

• **Staff:** Assists with scheduling, document uploading, client intake.

• **Client:** Views case status updates, scheduled appointments.

3. Main Processes:

1. Client registration and user profile management
2. Case file creation and status tracking
3. Appointment scheduling and hearing calendar management
4. Document uploading and versioning for cases
5. Billable hours tracking

4. Entities / Tables:

1. users
2. clients
3. cases
4. appointments
5. documents
6. time_entries

Part C — Complete ERD & Entity Specifications

1. Separated Entity Attributes & Detailed Structures

1. users Entity

Stores system accounts for administrators, lawyers, and staff members.

Attribute Name	Description / Data Type
user_id	Unique user identifier
username	System login username
email	User email address

password	Hashed account password
role	Role (Admin, Lawyer, Staff)
created_at	Timestamp of account creation

2. clients Entity

Stores information regarding individuals or companies represented by the law firm.

Attribute Name	Description / Data Type
client_id	Unique client identifier
user_id	Links to users table if client has portal access
fullname	Client's complete name or company name
phone	Contact phone number
email	Client email address
address	Physical mailing address

3. cases Entity

Stores legal case files managed by the firm.

Attribute Name	Description / Data Type
case_id	Unique case identifier
client_id	Reference to clients entity
assigned_lawyer_id	Reference to users entity (lawyer)
case_number	Official court or internal case docket number
title	Case name / subject matter
description	Summary details of the legal matter
status	Case status (Active, Pending, Closed)
filed_date	Date case was filed or opened

4. appointments Entity

Stores schedules for meetings, consultations, and court hearings.

Attribute Name	Description / Data Type
appointment_id	Unique appointment identifier
case_id	Reference to cases entity
client_id	Reference to clients entity
user_id	Reference to users entity (lawyer/staff)
title	Meeting or hearing title

appointment_date	Exact date and time of appointment
type	Consultation, Hearing, Client Meeting
status	Scheduled, Completed, Cancelled

5. documents Entity

Stores files, legal briefs, contracts, and court filings associated with cases.

Attribute Name	Description / Data Type
document_id	Unique document identifier
case_id	Reference to cases entity
uploaded_by	Reference to users entity
document_name	Original filename or document title
file_path	Server file storage path or URL
upload_date	Timestamp when file was uploaded

6. time_entries Entity

Stores billable work hours logged by lawyers against specific cases.

Attribute Name	Description / Data Type
time_entry_id	Unique time entry identifier
case_id	Reference to cases entity
user_id	Reference to users entity (lawyer)
hours	Number of billable hours logged
description	Work performed notes
date_logged	Date work was performed

2. Relationships & Cardinalities

The ERD utilizes strict relational cardinalities to maintain data integrity across firm operations:

• USERS (1) ■■■< CLIENTS (1:M): One user account can optionally link to a client portal profile.

• CLIENTS (1) ■■■< CASES (1:M): One client can have multiple legal cases; each case belongs strictly to one client.

• USERS/Lawyer (1) ■■■< CASES (1:M): One attorney can be assigned to manage multiple cases; each case has one lead assigned lawyer.

• CASES (1) ■■■< APPOINTMENTS (1:M): One case can have multiple scheduled meetings, consultations, or court hearings.

• CLIENTS (1) ■■■< APPOINTMENTS (1:M): Clients can have multiple appointments scheduled across their cases.

• CASES (1) ■■■< DOCUMENTS (1:M): One case can store multiple legal briefs, contracts, and court filings.

• CASES (1) ■■■< TIME_ENTRIES (1:M): Lawyers can log multiple billable hours entries against a specific case.

Part D — ERD Quality Check & Verification

✓ **Can every important piece of information be stored?** Yes. The 6 entities thoroughly capture client details, case notes, court calendars, document repositories, and billable hours.

✓ **Are the relationships logical?** Yes. Modeled closely after standard legal practice management software workflows.

✓ **Are cardinalities correct?** Yes. One-to-many (1:M) cardinalities accurately reflect operational realities.

✓ **Are there unnecessary duplicates?** No. Normalized structure prevents data duplication across entities.

✓ **Can the database support actual tables?** Yes. Fully compatible with relational database management systems (RDBMS) such as PostgreSQL, MySQL, and SQL Server.